Shane McGrath

Senior Frontend Engineer

+44 7493 370066 | shane.mcgrath2@gmail.com | LinkedIn | GitHub | Portfolio

Summary

Senior Frontend Engineer with 8+ years of experience building and evolving production web applications with TypeScript, React and GraphQL. Experienced in owning complex frontend systems, establishing reusable UI architecture, integrating distributed services and supporting reliable production environments. Strong technical collaborator and mentor with experience solving ambiguous problems and delivering measurable improvements to product quality, developer productivity and customer experience.

Experience

Senior Software Engineer – Sustain.Life Aug. 2022 – Present
Enterprise carbon accounting platform built with React, TypeScript and GraphQL. *Remote, Ireland*

- Owned and developed enterprise-scale frontend application using **React, TypeScript and GraphQL** within a large monorepo, driving architecture and implementation across complex product areas.
- Designed reusable frontend components and workflows with UI/UX, establishing consistent patterns through **Storybook** and reducing duplicated implementation across the product.
- Improved frontend reliability and development velocity by expanding automated test coverage, reducing daily manual QA effort by **45%**.
- Built end-to-end integrations with third-party providers including **Codat, Utility API and Arcadia**, supporting key product workflows.
- Worked across frontend and backend services in a distributed system using **microservices and event sourcing**.
- Owned production reliability and zero-downtime deployments, troubleshooting CI/CD pipelines, Kubernetes resources and critical incidents through on-call support.
- Mentored engineers through pair programming, technical guidance and code reviews, helping establish maintainable engineering practices across the team.
- Translated ambiguous customer and product requirements into maintainable technical solutions with a focus on product outcomes.

Frontend Developer – Axial3D Aug. 2018 – July 2022
Medical technology startup delivering 3D printing and machine-learning workflows. *Belfast, NI*

- Built responsive frontend applications for displaying **medical image segmentations, 3D visualizations and annotations**, translating Figma designs into production workflows.
- Built interactive **3D medical visualization** workflows using **Three.js and WebGL**, enabling users to inspect and interact with machine-learning-generated medical imaging data in the browser.
- Reduced customer onboarding time by **35%** through usability improvements and redesign of the sign-up workflow.
- Rapidly delivered new product capabilities using customer analytics and feedback to validate product direction and prioritize engineering work.
- Designed and implemented **REST APIs** supporting customer workflows around medical imaging data in accordance with the **DICOM** standard.
- Implemented frontend automated testing using **Cypress and Jest**, improving regression coverage and confidence in production deployments.
- Applied software design patterns, SDLC practices, Git-based version control and testing strategies to build maintainable production software.
- Onboarded new engineers into the development environment, sharing technical knowledge, system context and development practices.
- Earned the **AWS Cloud Practitioner** certification, developing practical knowledge of cloud architecture, security and AWS services.

Design Engineer Intern – PMD Solutions Apr. 2017 – June 2018
Contributed to product design for a sleep apnoea wearable device and companion application. *Cork, Ireland*

- Designed mechanical components in **SolidWorks** and developed rapid-prototyping workflows using 3D printing, including training team members in 3D printer usage.
- Conducted user research and collaborated with designers and developers on web and native mobile product development, contributing to roadmap decisions.

Education

University of Galway — Ireland
Higher Diploma Software Design and Development — *1 year*

University of Galway — Ireland
Bachelor of Engineering (Mechanical Engineering) — *4 years*

Technical Skills

Languages: TypeScript, JavaScript, HTML, CSS, SQL
Frontend: React, Next.js, Vue, Svelte, Angular, Three.js, WebGL, Material UI
Testing: Jest, Vitest, Cypress, Selenium
Backend & APIs: Node.js, GraphQL, REST, tRPC
Cloud & Infrastructure: AWS, Azure, Docker, Kubernetes, CI/CD
Tools: Git, Storybook, Vite, Webpack, Postman, Claude Code, Sentry, Figma, Jira